

CHALLENGE 03

Game Load Time — 6–8 Seconds to Near-Instant

Empire of Gold

Instant game launch, without touching the certified bundle

5,430 ms

today, cold

243 ms

with our prefetch

22×

faster, measured

0 bytes

changed in the game

Time to Play screen · 8 Mbps / 80 ms · two runs per scenario, each in a fresh browser profile

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THE PROBLEM

Slow loads do not just cost seconds. They cost the catalogue.

A player taps a tile and waits for a **56.3 MB** certified bundle.

- **Measured:** 5.4 s to the Play screen on an ordinary mobile link — then ~45 MB more streams in behind it while they play.
- **The second-order cost is worse.** In a product whose core loop is choosing and switching games, a 6–8 s penalty on every tap teaches players not to explore. They settle into a familiar few, and new content never gets a fair chance.

THE CONSTRAINT IS STRUCTURAL

Games are certified third-party bundles, contractually fixed. Every obvious fix is closed:

- ✗ Cannot modify the bundle — it is certified.
- ✗ Cannot move it to a CDN host — a service worker cannot see across origins.
- ✗ Cannot push 56 MB at everyone — ~95% of lobby visitors never open a game.

So the solution space is everything around the game.

THE SOLUTION

Move the bytes while they browse, not after they tap.

A service worker warms the shared bundle on the lobby, in three strictly ordered passes.

1 SLICE

6.8 MB

The 27 files that gate the Play screen
every visitor

2 AUDIO

10.7 MB

Sound the game requests during boot
every visitor, after pass 1

3 THE REST

38.8 MB

Spines, pay table, free-spins loop
returning players only

One bundle, thirty games

Every title boots the same certified build, so prefetch accuracy is 100% by construction — there is no wrong game to guess, and trying a new one costs the player nothing.

The worker only serves

It never rewrites or injects. The bundle stays byte-identical, so nothing needs re-certifying — and speed can never bypass a limit, check or age gate.

243 ms, against a 500 ms target.

TIME TO PLAY SCREEN · 8 Mbps / 80 ms

Cold, no prefetch **5,430 ms**

Opened while prefetch still running **2,769 ms**

Opened after prefetch finished **243 ms**

A prefetch in flight halves the wait rather than competing with it — it is already pulling the files that game needs.

AGAINST THE CHALLENGE METRICS

Cold load < 500 ms **243 ms**

Cache hit / prefetch accuracy **100%**

Fully loaded game **51 s → 0.8 s**

Cold-load p50 / p95 **not yet aggregated**

We measure every load; nothing aggregates them yet. Percentiles need production traffic.

On a warm open: 141 bundle requests, 140 of them transfer nothing.

That is a property of the cache, not of the link — it reproduces on any device. All of the gain is raw speed: no spinner, no skeleton screen, no perceived-loading trick.

BUSINESS IMPACT

€0.24 per 1,000 sessions.
Break-even at a 0.16% lift.

THE COST

- **17.5 MB** per lobby session — €0.35 per 1,000 at €0.02/GB.
- **Most is not new spend.** For the 30% who open a game, these are bytes they would have downloaded anyway. Additional: €0.24 per 1,000.
- **Integration:** no change to the game, no re-certification, no backend. Rollback is deleting one line.

THE VALUE

- **27.5 seconds** of dead waiting removed per playing session — 2.3 hours per 1,000 lobby sessions.
- **The bar is low:** if a playing session is worth €0.50, this pays for itself at a 0.16% lift. Even at 5× the bandwidth rate it stays under 1%.
- **Gated on your own data:** a cold visitor opens a game 4.5% of the time; a returning one, 46.7%. The 38.8 MB pass goes only past that line.

What we would need to close the analysis

Your real egress rate · your value per playing session · the same cohort analysis on full traffic.

A two-week experiment at 5% of traffic produces all three. Turning it off is deleting one line.